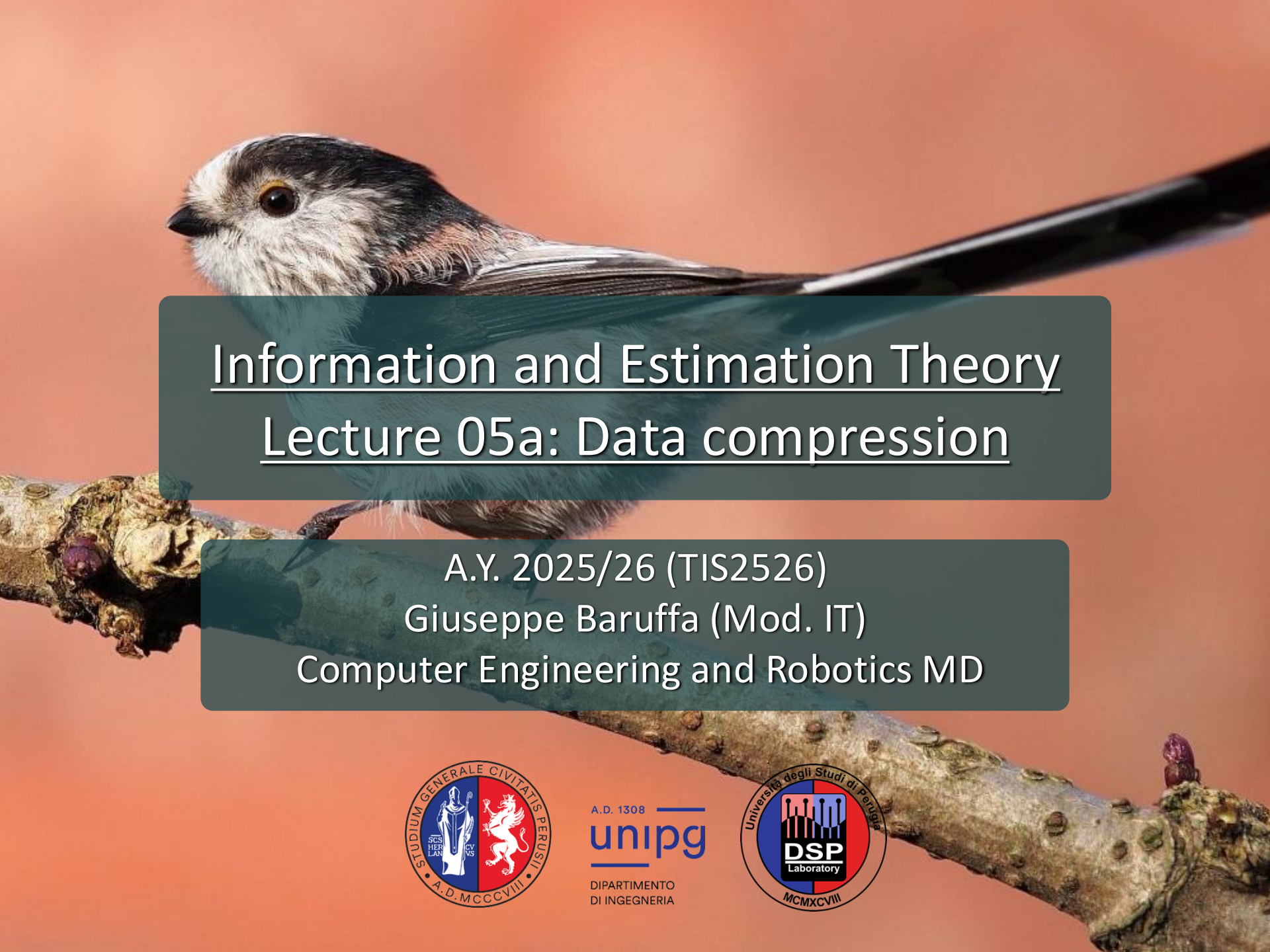


Information and Estimation Theory
Lecture 05a: Data compression

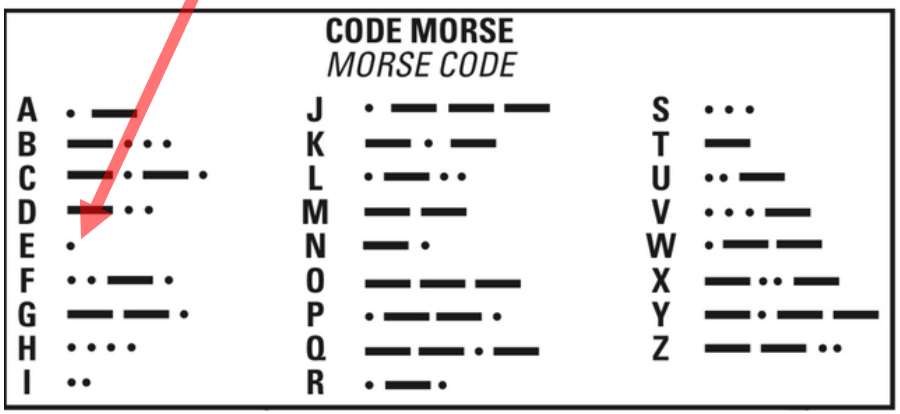
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Data Compression

- We will establish the fundamental limit for the compression of information, a need in advanced communications
- **Data compression** can be achieved by assigning short descriptions to the most frequent outcomes of the data source, and (necessarily) longer descriptions to the less frequent ones
  - E.g., in Morse code, the most frequent symbol is represented by a single dot





Data Compression

- We want to find the **shortest average description length** of a RV
- We define an instantaneous code and prove the Kraft inequality, which asserts that the codeword lengths must look like a PMF
- Elementary calculus shows that the expected description length must be greater than or equal to the entropy
- Shannon's simple construction shows that the expected description length can achieve this bound asymptotically
- Entropy sets as a natural measure of efficient description length
  - We will show that Huffman coding is competitively optimal
- Thus, **entropy is the data compression limit**
  - Codes achieving *H* turn out to be optimal from many points of view

Source Codes

- A **source code (SC)** *C* for a RV *X* is a mapping from the range $\mathcal{X}$ to the set $\mathcal{D}^*$ (*codebook*) of finite-length strings (*codewords*, CW) of symbols from a *D*-ary alphabet $\mathcal{D}$
- Let *C*(*x*) denote the CW corresponding to *x* and let *l*(*x*) denote the length of *C*(*x*)
  - E.g., C(*red*) = 00, C(*blue*)= 11 is a SC for $\mathcal{X}$ = {*red*,*blue*} with alphabet $\mathcal{D}$={0,1} and codebook $\mathcal{D}^*$ = {00,11}. The lengths are *l*(*red*) = *l*(*blue*) = 2

- The **expected length *L*(*C*)** of a SC *C*(*X*) for a RV *X* with PMF *p*(*x*) is

$$L(C) = \sum_{x \in \mathcal{X}} p(x) l(x)$$

  - Without loss of generality, we assume the *D*-ary alphabet $\mathcal{D}$ = {0,1,…,*D*−1}

Example

- Let X be a RV with the following PMF and SC

$$\Pr\{X=1\}=\frac{1}{2} \qquad \text{codeword } C(1)=0$$
$$\Pr\{X=2\}=\frac{1}{4} \qquad \text{codeword } C(2)=10$$
$$\Pr\{X=3\}=\frac{1}{8} \qquad \text{codeword } C(3)=110$$
$$\Pr\{X=4\}=\frac{1}{8} \qquad \text{codeword } C(4)=111$$

- The entropy $H(X)$ of X is $H(X)=-\frac{1}{2}\log\frac{1}{2}-\frac{1}{4}\log\frac{1}{4}-2\frac{1}{8}\log\frac{1}{8}=\frac{7}{4}=1.75$ bits, and the expected length $L(C)=\mathrm{E}\{l(X)\}$ of this code is

$$L\big(C(X)\big)=\frac{1}{2}1+\frac{1}{4}2+2\frac{1}{8}3=\frac{7}{4}=1.75$$

bits

- This code has the same average length as the entropy
- We note that any sequence of bits can be uniquely decoded into a sequence of symbols of X
  - E.g., the bit string 0110111100110 is decoded as 134213

Example

- Another simple example of a SC for a RV

$$\Pr\{X=1\}=\frac{1}{3} \quad \text{codeword } C(1)=0$$
$$\Pr\{X=2\}=\frac{1}{3} \quad \text{codeword } C(2)=10$$
$$\Pr\{X=3\}=\frac{1}{3} \quad \text{codeword } C(3)=11$$

- This code as well is uniquely decodable
  - E.g., 101110011 decodes into 23213
- However, in this case the entropy is

$$H(X)=-3\frac{1}{3}\log\frac{1}{3}=\log 3=1.58$$

bits and the average length of the encoding is

$$L\big(C(X)\big)=\frac{1}{3}1+2\frac{1}{3}2=\frac{5}{3}=1.67$$

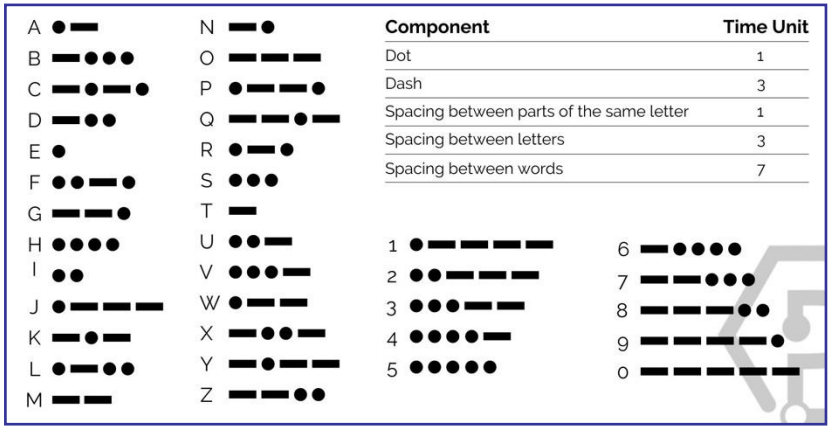
bits, thus E{$l(X)$} > $H(X)$

Example: Morse Code

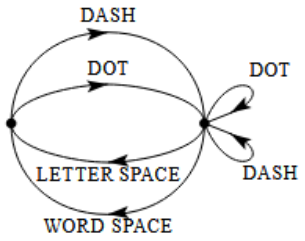
- The Morse code is a reasonably efficient code for the English alphabet $\mathcal{X}$ = {A,B,C,…,Z} using symbols from a quaternary (D=4) alphabet $\mathcal{D}$ = {• (dot), — (dash), _ (letter space), _ (word space)}
  - International standard: *Recommendation ITU-R M.1677-1*
- Short CWs are used for frequent letters (e.g., "•" represents "E") and long CWs for infrequent letters (e.g., "Q" is "——•—")
- E.g., "THE CAR" becomes

— _ •••• _ • _ —•—• _ •— _ •—•

T H E C A R



- Morse code is not optimal
  - CWs for letters do not contain spaces (except the ending space) and no space can follow another space
  - Shannon (1948) noticed this Markovity



Nonsingular Codes

- A code is said to be **nonsingular** if every element of the range of *X* maps into a different string in $\mathcal{D}^*$

$$\forall\, x \neq x' \quad \Rightarrow \quad C(x) \neq C(x')$$

  - E.g., if *C*(*green*)=11, *C*(*red*)=010 and *C*(*blue*)=010, code is singular
- Nonsingularity suffices for an unambiguous description of a single value of *X*
- But we usually wish to send a sequence of values of *X*
- In such cases we can ensure decodability by adding a special symbol (a "*comma*") between any two CWs
- This is an inefficient use of the special symbol: better to develop "selfpunctuating" or instantaneous codes

Code Extension

- The **extension** C^* of a code C is the mapping from finite length strings of $\mathcal{X}$ to finite-length strings of $\mathcal{D}$, defined by

$$C(x_1 x_2 \cdots x_n) = C(x_1)C(x_2)\cdots C(x_n)$$

where $C(x_1)C(x_2)\cdots C(x_n)$ indicates concatenation of the corresponding CWs

  - E.g., if $C(x_1) = 00$ and $C(x_2) = 11$, then $C(x_1x_2) = C(x_1)C(x_2) = 0011$

Uniquely Decodable Codes

- A code is called **uniquely decodable** if its extension is nonsingular

$$\forall\, x_1 x_2 \dots x_n \neq x_1' x_2' \dots x_n' \quad \Rightarrow \quad C(x_1 x_2 \dots x_n) \neq C(x_1' x_2' \dots x_n')$$

- In other words, any encoded string in a uniquely decodable code has only one possible source string producing it
- However, one may have to look at the entire string to determine even the first symbol in the corresponding source string

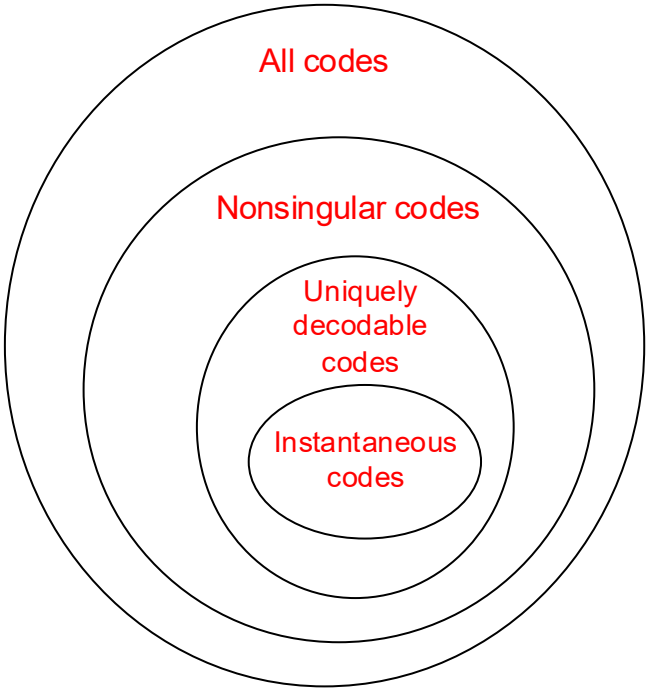
Prefix Codes

- A code is called a **prefix code** or an **instantaneous code** if no CW is a prefix of any other CW
  - Prefix in the sense of prefix-free
- An instantaneous code can be decoded without reference to future CWs since the end of a CW is immediately recognizable
- Hence, for an instantaneous code, the *i*th symbol x_i can be decoded as we come to the end of the CW corresponding to it
  - We need not wait to see the upcoming CWs
- An instantaneous code is selfpunctuating:
  we can browse the sequence of code symbols and add the commas to separate the CWs without looking at later symbols
  - E.g., the binary string 01011111010 produced by the previous example code is parsed as 0,10,111,110,10

Example: Classes of Codes

X	Singular	Nonsingular, but not uniquely decodable	Uniquely decodable, but not instantaneous	Instantaneous
1	0	0	10	0
2	0	010	00	10
3	0	01	11	110
4	0	10	110	111

- For the nonsingular code, the CW 010 has three possible source sequences: 2 (010) or 14 (0,10) or 31 (01,0)
  - Not uniquely decodable



Example: Classes of Codes

- The uniquely decodable SC is not prefix-free and not instantaneous
- If the first two bits are 00 or 10, decode them immediately
- If the first two bits are 11, we must look at the following bits
  - If the next bit is a 1, the first source symbol is a 3 1110
  - If the number of 0's immediately following the 11 is odd, the first CW must be 110 and the first source symbol must be 4 1100010
  - If the number of 0's is even, the first source symbol is a 3 110010
- By repeating, we can see this is uniquely decodable
- Obvious: the last code is instantaneous since no codeword is a prefix of any other

X	Singu-lar	Nonsingular, but not uniquely decodable	Uniquely decodable, but not instantaneous	Instanta-neous
1	0	0	10	0
2	0	010	00	10
3	0	01	11	110
4	0	10	110	111

Kraft Inequality

- We wish to construct instantaneous codes of minimum expected length to describe a given source
- We cannot assign short CWs to all source symbols and still be prefix-free
- The set of CW lengths possible for instantaneous codes is limited

- **Theorem (Kraft inequality):** For any instantaneous code (prefix code) over an alphabet of size D, the codeword lengths $l_1, l_2, \ldots, l_m$ must satisfy the inequality

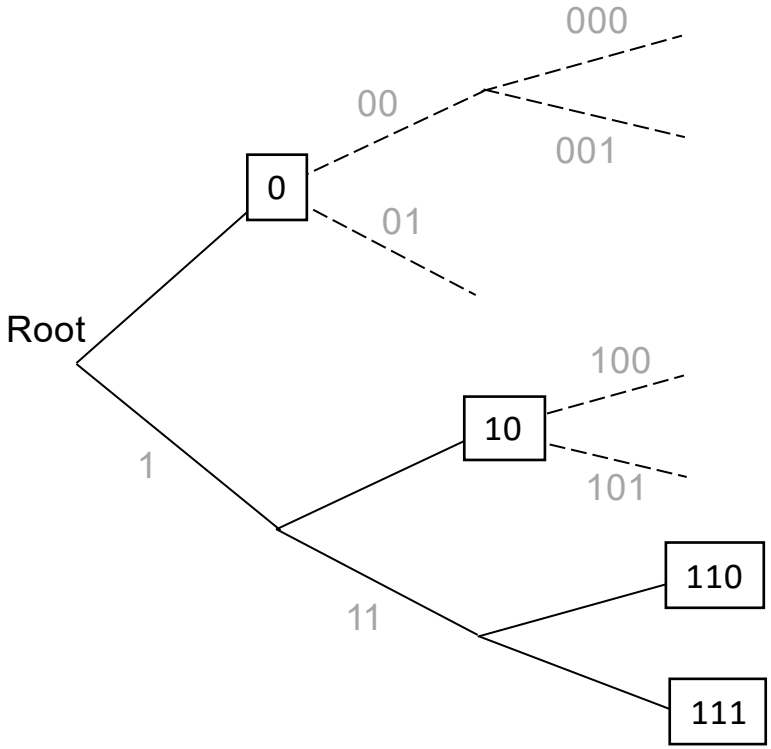
$$\sum_{i=1}^{m} D^{-l_i} \leq 1$$

Conversely, given a set of CW lengths that satisfy this inequality, there exists an instantaneous code with these word lengths

Kraft Inequality (proof)

- *Proof:* Consider a *D*-ary tree in which each node has *D* children and the branches do not fuse among themselves
- Let the branches of the tree represent the symbols of the CW
  - E.g., the *D* branches arising from the root node represent the *D* possible values of the first symbol of the CW

- Then each CW is a leaf on the tree
- The path from the root traces out the symbols of the CW
- The prefix condition on the CWs implies that no CW is an ancestor of any other CW on the tree
- Hence, each CW eliminates its descendants as possible CWs



Kraft Inequality (proof cont'd)

- Let l_{max} be the length of the longest CW of the set of CWs
- Consider all nodes of the tree at level l_{max}
- Some are CWs, some are CWs descendants, and some are neither
- The CW $C(x_i)$ of length l_i has a real leaf at level l_i , generating an offspring of $D^{l\max - l_i}$ descendants at level l_{max}
  - E.g., if l_{max} =5 and D=2, each one of the CW of length 3 (l_i=3) has $2^{5-3}=2^2=4$ potential descendants at level l_{max} =5

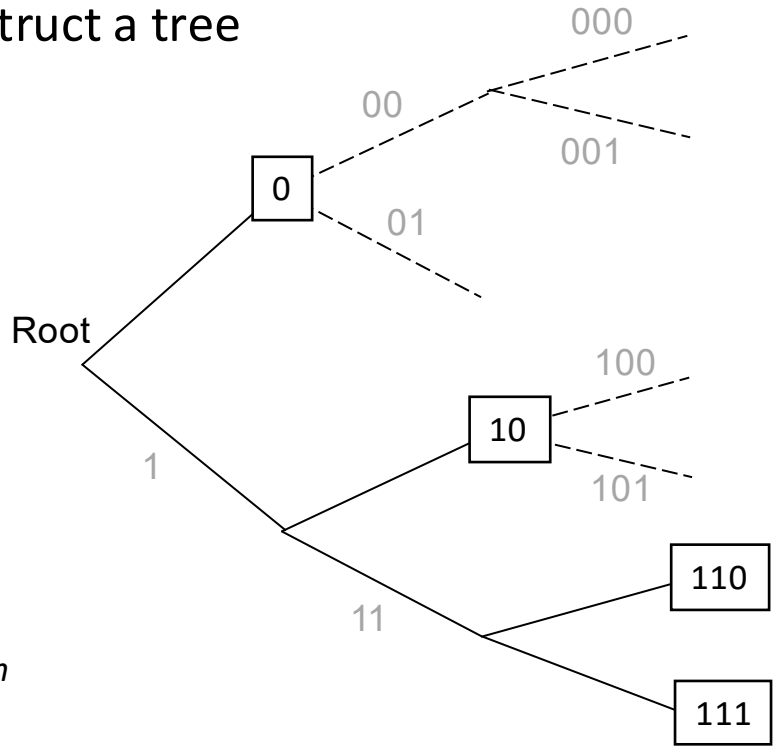
- Each of these descendant sets is disjoint, per the construction of the tree
- Also, the total number of nodes in these sets must be less than or equal to the number of leaves $D^{l\max}$ at the final level
- Hence, summing over all the m codewords, we find

$$\sum_{i=1}^{m} D^{l\max - l_i} \leq D^{l\max} \qquad D^{l\max} \sum_{i=1}^{m} D^{-l_i} \leq D^{l\max} \qquad \sum_{i} D^{-l_i} \leq 1$$

Kraft Inequality (proof cont'd)

- Theorems in IT are often proved in two steps
  1. Demonstrate achievability of the claimed property
  2. Prove the converse, i.e., no other solution exists with the claimed property
- E.g., if (direct) $a > b$ then (converse) $b < a$

- So, conversely, given any set of CW lengths $l_1, l_2, \ldots, l_m$ that satisfy the Kraft inequality, we can always construct a tree
- Label the first node (lexicographically) of depth l_1 as CW 1, and remove its descendants from the tree
- Then label the first remaining node of depth l_2 as CW 2, and remove its descendants
- …
- Proceeding this way, we construct a prefix code with the specified $l_1, l_2, \ldots, l_m$



Extended Kraft Inequality

- **Theorem:** For any countably infinite set of CWs that form a prefix code, the CW lengths satisfy the extended Kraft inequality

$$\sum_{i=1}^{\infty} D^{-l_i} \leq 1$$

Conversely, given any $l_1, l_2, \ldots$ satisfying the extended Kraft inequality, we can construct a prefix code with these CW lengths

  - *Proof:* Let the D-ary alphabet be $\{0,1,\ldots,D-1\}$. Consider the ith codeword $y_1 y_2 \cdots y_{l_i}$. Let $0.y_1 y_2 \cdots y_{l_i\ D}$ be the D-ary form of the real number given by the D-ary expansion

$$0.y_1 y_2 \cdots y_{l_i\ D} = \sum_{j=1}^{l_i} y_j D^{-j}$$

  - This CW corresponds to the interval $\left[0.y_1 y_2 \cdots y_{l_i\ D}, 0.y_1 y_2 \cdots y_{l_i\ D} + \frac{1}{D^{l_i}}\right)$, the set of all real numbers whose D-ary expansion begins with $0.y_1 y_2 \cdots y_{l_i}$

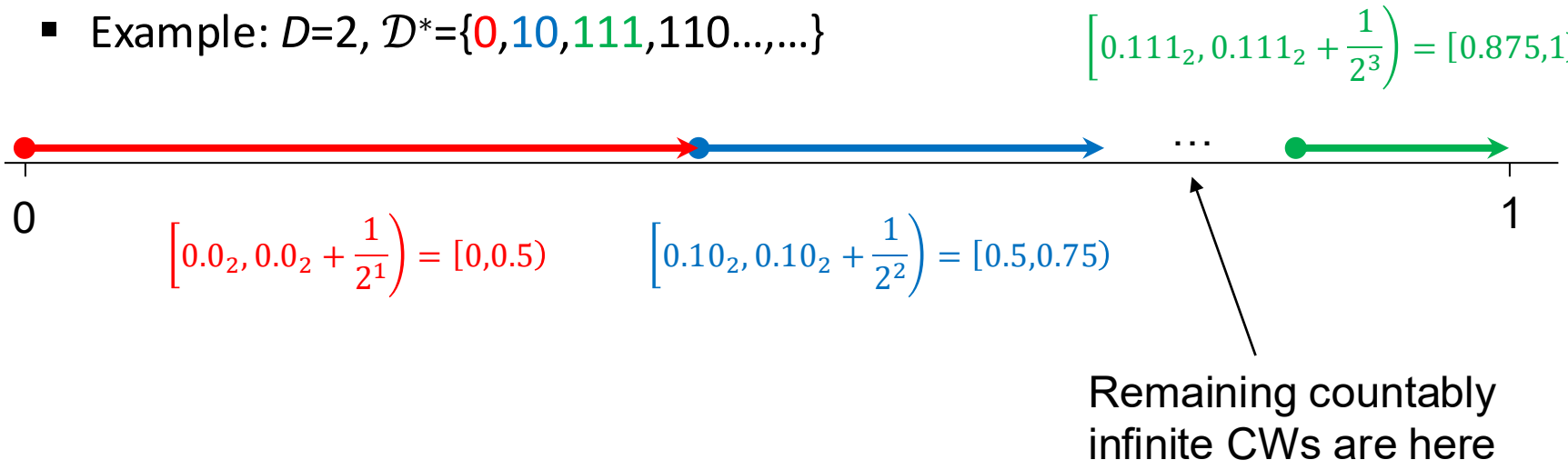
Extended Kraft Inequality (proof cont'd)

- This is a subinterval of the unit interval [0, 1]. By the prefix condition, these intervals are disjoint. Hence, the sum of their lengths must be less than or equal to 1. This proves that

$$\sum_{i=1}^{\infty} D^{-l_i} \leq 1$$

- Converse: just as in the finite case, we can reverse the proof to construct the code for a given $l_1, l_2, \ldots$ that satisfies the Kraft inequality

- Example: $D=2$, $\mathcal{D}^*=\{0,10,111,110\ldots,\ldots\}$



Optimal Codes

- We proved that any CW set that satisfies the prefix condition **must** satisfy the Kraft inequality, and that the Kraft inequality is a **sufficient** condition for the existence of a CW set with the specified set of CW lengths
- We now consider the problem of **finding the prefix code with the minimum expected length**
- This is equivalent to finding the set of lengths $l_1,l_2,\ldots,l_m$ satisfying the Kraft inequality with expected length $L = \Sigma_i p_i l_i$ less than the expected length of any other prefix code
  - This is a standard integer optimization problem:

    Minimize $L = \sum_i p_i l_i$ over all integers $l_1,l_2,\ldots,l_m$ satisfying $\sum_i D^{-l_i} \leq 1$

$$\min_{l_1,\ldots,l_m \in \mathbb{N}} \sum_{i=1}^{m} p_i l_i$$

$$s.t. \sum_{i=1}^{m} D^{-l_i} \leq 1$$

Optimal Codes

- A simple analysis suggests the form of the minimizing l_i^*
- Neglect the integer constraint on l_i and assume equality
- Write the constrained minimization using Lagrange multipliers as the minimization of

$$J = \sum_{j=1}^{m} p_j l_j + \lambda \left(\sum_{j=1}^{m} D^{-l_j} \right)$$

- Differentiating with respect to l_i, we obtain

$$\frac{\partial J}{\partial l_i} = p_i - \lambda D^{-l_i} \ln D$$

$\frac{d}{dx} a^x = a^x \ln a$

- Setting the derivative to 0, we obtain

$$D^{-l_i} = \frac{p_i}{\lambda \ln D}$$

Optimal Codes

- Substituting this in the constraint we find

$$\sum_{j=1}^{m} D^{-l_j} = \sum_{j=1}^{m} \frac{p_j}{\lambda \ln D} = \frac{1}{\lambda \ln D} \sum_{j=1}^{m} p_j = \frac{1}{\lambda \ln D} \xrightarrow{\text{constraint}} \frac{1}{\lambda \ln D} = 1 \;\rightarrow\; \lambda = \frac{1}{\ln D}$$

giving $p_i = D^{-l_i}$ and the optimal code lengths

$$l_i^* = -\log_D p_i$$

- This noninteger choice of CW lengths yields an expected length

$$L^* = \sum_{i=1}^{m} p_i l_i^* = -\sum_{i=1}^{m} p_i \log_D p_i = H_D(X) = H(X) \log_D 2$$

- Since l_i must be integer, we will not be able to set the CW lengths
- Instead, we should choose lengths l_i "close" to the optimal set

Theorem on Optimal Codes

- **Theorem:** The expected length L of any instantaneous D-ary code for a RV X is greater than or equal to the entropy $H_D(X)$

$$L \geq H_D(X)$$

with equality if and only if $D^{-l_i} = p_i$

  - *Proof:* difference between the expected length and the entropy

$$\begin{aligned} L - H_D(X) &= \sum_i p_i l_i - \sum_i p_i \log_D \frac{1}{p_i} = -\sum_i p_i \log_D D^{-l_i} + \sum_i p_i \log_D p_i \\ &= \sum_i p_i \log_D \frac{p_i}{D^{-l_i}} = \sum_i p_i \log_D \frac{p_i}{r_i \sum_j D^{-l_j}} = \sum_i p_i \log_D \frac{p_i}{r_i} - \log_D \sum_i D^{-l_i} \\ &= D(p \parallel r) + \log_D \frac{1}{\sum_i D^{-l_i}} \geq 0 \end{aligned}$$

by letting $r_i = D^{-l_i}/\sum_j D^{-l_j}$ denote a PMF generated by CW lengths, by the nonnegativity of relative entropy $D(p\|r)$ and by $\sum_i D^{-l_i} \leq 1$ (Kraft inequality). Hence, $L \geq H$ with equality iff $p_i = D^{-l_i}$, i.e., iff $-\log_D(p_i)$ is an integer for all i

D-adic Distributions

- A PMF is called ***D*-adic** if each probability is D^{-n} for some n
- Thus, we have equality in the theorem iff the PMF of X is D-adic
- The preceding proof also indicates a procedure for finding an optimal code
  - Find the D-adic PMF that is closest (in the relative entropy sense) to the PMF of X
- This distribution provides the set of CW lengths
- Construct the code by choosing the 1st available node as in the proof of the Kraft inequality, to have an optimal code for X
- However, this procedure is not easy, since the search for the closest D-adic distribution is not obvious
- Shortly, we will describe a good suboptimal procedure (Shannon–Fano coding) and a simple procedure (Huffman coding) for finding the optimal code

Bounds on Code Length

- We show a code that achieves an expected description length L within 1 bit of the lower bound

$$H(X) \leq L < H(X) + 1$$

- The constrained minimization showed that the optimal CW lengths can be found by the D-adic PMF closest to the PMF of X in relative entropy, that is, by finding the D-adic $r(x)$, $r_i = D^{-l_i}/\sum_j D^{-l_j}$, minimizing

$$L - H_D = D(p \parallel r) - \log\left(\sum_i D^{-l_i}\right) \geq 0$$

- The choice of codeword lengths $l_i = \log_D(1/p_i)$ yields $L = H$
- Since $\log_D(1/p_i)$ may not be integer, we round it up with the ceiling function

$$l_i = \left\lceil \log_D \frac{1}{p_i} \right\rceil$$

Bounds on Code Length

- These lengths satisfy the Kraft inequality, since

$$\sum_i D^{-l_i} = \sum_i D^{-\left\lceil \log_D \frac{1}{p_i} \right\rceil} \leq \sum_i D^{-\log_D \frac{1}{p_i}} = \sum_i p_i = 1$$

$0 \leq p_i \leq 1, \qquad D > 1$

$\frac{1}{p_i} \geq 1$

$\log_D \frac{1}{p_i} \geq 0$

$\left\lceil \log_D \frac{1}{p_i} \right\rceil \geq \log_D \frac{1}{p_i}$

$-\left\lceil \log_D \frac{1}{p_i} \right\rceil \leq -\log_D \frac{1}{p_i}$

$D^{-\left\lceil \log_D \frac{1}{p_i} \right\rceil} \leq D^{-\log_D \frac{1}{p_i}}$

- This choice of codeword lengths satisfies

$$\log_D \frac{1}{p_i} \leq l_i < \log_D \frac{1}{p_i} + 1$$

- Multiplying by p_i and summing over i, we obtain

$$H_D(X) \leq L < H_D(X) + 1$$

$p_i \log_D \frac{1}{p_i} \leq p_i l_i < p_i \log_D \frac{1}{p_i} + p_i$

$\sum_i p_i \log_D \frac{1}{p_i} \leq \sum_i p_i l_i < \sum_i p_i \log_D \frac{1}{p_i} + \sum_i p_i$

Theorem on Optimal Codes Length

- **Theorem:** Let $l_1^*, l_2^*, \ldots, l_m^*$ be optimal CW lengths for a source PMF $p(x)$ and a D-ary alphabet and let $L^* = \Sigma_i p_i l_i^*$ be the associated expected length of an optimal code. Then

$$H_D(X) \leq L^* < H_D(X) + 1$$

  - *Proof:* Let $l_i = \lceil \log_D(1/p_i) \rceil$. Then l_i satisfies the Kraft inequality

$$H_D(X) \leq L = \sum_i p_i l_i < H_D(X) + 1$$

  But since $L^* \leq L = \Sigma_i p_i l_i$, and since $L^* \geq H_D$, we have

$$H_D(X) \leq L^* \leq L < H_D(X) + 1$$

Lowering the Bounds

- There is an overhead per symbol that is at most 1 bit, since $\log(1/p_i)$ is not always an integer
  - We can reduce it by spreading it out over many symbols
- Let us consider a system in which we send a sequence of n symbols from X i.i.d. $\sim p(x)$
- Consider the n symbols as a supersymbol from the alphabet $\mathcal{X}^n$
- Define L_n to be the expected CW length per input symbol, that is, if $l(x^n)$ is the length of the binary CW associated with $x^n=(x_1,x_2,\ldots,x_n)$, then

$$L_n = \frac{1}{n}\sum_{x^n \in \mathcal{X}^n} p(x^n) l(x^n) = \frac{1}{n}\mathrm{E}\{l(X^n)\}$$

Lowering the Bounds

- We can now apply the bounds derived above to the code

$$H(X^n) \leq \mathrm{E}\{l(X^n)\} < H(X^n) + 1$$

- Since $X_1, X_2, \ldots, X_n$ are i.i.d. $\sim p(x)$, $H(X_1, X_2, \ldots, X_n) = \Sigma_i H(X_i) = nH(X)$
- Dividing by n, we obtain

$$H(X) \leq L_n < H(X) + \frac{1}{n}$$

- Hence, by using large block lengths we can achieve an expected codelength per symbol **arbitrarily close to the entropy**

Lowering the Bounds

- We can use the same argument for a sequence of symbols from a SP that is not necessarily i.i.d.
- In this case, we still have the bound

$$H(X_1, X_2, \dots, X_n) \leq \mathrm{E}\{l(X_1, X_2, \dots, X_n)\} < H(X_1, X_2, \dots, X_n) + 1$$

- Dividing by *n* again we obtain

$$\frac{H(X_1, X_2, \dots, X_n)}{n} \leq L_n < \frac{H(X_1, X_2, \dots, X_n)}{n} + \frac{1}{n}$$

- If the SP is stationary, then $\mathrm{H}(X_1,X_2,\dots,X_n)/n \to H(\mathcal{X})$, and the **expected description length tends to the entropy rate** as $n \to \infty$

Theorem on Expected Length

- **Theorem:** The minimum expected CW length per symbol satisfies

$$\frac{H(X^n)}{n} \leq L_n^* < \frac{H(X^n)}{n} + \frac{1}{n}$$

Moreover, if $X_1, X_2, \ldots, X_n$ is a stationary SP,

$$\lim_{n\to\infty} L_n^* = H(\mathcal{X})$$

where $H(\mathcal{X})$ is the entropy rate of the SP

- The theorem provides another justification for the definition of entropy rate:

"the expected number of bits per symbol required to describe the SP"

Wrong Code

- What happens to the expected description length if the code is designed for the wrong PMF?
  - E.g., the wrong PMF may be the best estimate that we can make of the unknown true PMF
- We consider the **Shannon code assignment** $l(x) = \log[1/q(x)]$, designed for the PMF $q(x)$, while the true PMF is $p(x)$
  - Thus, we will not achieve the expected length $L \approx H(p) = -\sum_x p(x) \log p(x)$
- We will show that the increase in expected description length due to the incorrect distribution is the relative entropy $D(p\|q)$
- Thus, $D(p\|q)$ has a concrete interpretation as the **increase in descriptive complexity** due to incorrect information

Theorem on Wrong Code

- **Theorem:** The expected length under $p(x)$ of the code assignment $l(x)=\log[1/q(x)]$ satisfies

$$H(p) + D(p \parallel q) \leq \mathrm{E}_p\{l(X)\} < H(p) + D(p \parallel q) + 1$$

or

$$H(p \parallel q) \leq \mathrm{E}_p\{l(X)\} < H(p \parallel q) + 1$$

  - *Proof:* the expected length is (the lower bound can be derived similarly)

$$\mathrm{E}\{l(X)\} = \sum_x p(x) \left\lceil \log \frac{1}{q(x)} \right\rceil < \sum_x p(x) \left(\log \frac{1}{q(x)} + 1 \right)$$
$$= \sum_x p(x) \log \frac{1}{q(x)} + 1 = H(p \parallel q) + 1 = D(p \parallel q) + H(p) + 1 = H(p \parallel q) + 1$$

- Thus, believing that the PMF is $q(x)$ when the true PMF is $p(x)$ **incurs a penalty** of $D(p\|q)$ **in the average description length**

Kraft and Uniquely Decodable Codes

- Any instantaneous code must satisfy the Kraft inequality
- The class of uniquely decodable codes is larger than the class of instantaneous codes, so one expects to achieve a lower expected CW length if L is minimized over all uniquely decodable codes
- We will prove that the class of uniquely decodable codes does not offer any further possibilities for the set of CW lengths than do instantaneous codes

McMillan's Theorem

- **Theorem:** The CW lengths of any uniquely decodable *D*-ary code must satisfy the Kraft inequality

$$\sum_i D^{-l_i} \leq 1$$

Conversely, given a set of CW lengths that satisfy this, we can construct a uniquely decodable code with these CW lengths

- *Proof:* Consider C^k, the *k*th extension of the code, formed by the concatenation of *k* repetitions of the given uniquely decodable code C.
- By the definition of unique decodability, the *k*th extension of the code is nonsingular.
- Since there are only D^n different *D*-ary strings of length *n*, unique decodability implies that the number of code sequences of length *n* in the *k*th extension of the code must be no greater than D^n

McMillan's Theorem (proof cont'd)

- Let the CW lengths of the symbols $x \in \mathcal{X}$ be denoted by $l(x)$
- For the extension code, the length of the code sequence is

$$l(x_1, x_2, \ldots, x_k) = l(x^k) = \sum_{i=1}^{k} l(x_i)$$

and the inequality to prove is

$$\sum_{x \in \mathcal{X}} D^{-l(x)} \leq 1$$

- The trick is to consider the kth power of this quantity. Thus,

$$\left(\sum_{x \in \mathcal{X}} D^{-l(x)} \right)^k = \sum_{x_1 \in \mathcal{X}} \sum_{x_2 \in \mathcal{X}} \ldots \sum_{x_k \in \mathcal{X}} D^{-l(x_1)} D^{-l(x_2)} \ldots D^{-l(x_k)}$$

$$= \sum_{x_1, x_2, \ldots, x_k \in \mathcal{X}^k} D^{-l(x_1)} D^{-l(x_2)} \ldots D^{-l(x_k)} = \sum_{x_1, x_2, \ldots, x_k \in \mathcal{X}^k} D^{-\sum_{i=1}^{k} l(x_i)}$$

$$= \sum_{x^k \in \mathcal{X}^k} D^{-l(x^k)}$$

McMillan's theorem (proof cont'd)

- Gather the terms by word lengths to obtain

$$\sum_{x^k \in \mathcal{X}^k} D^{-l(x^k)} = \sum_{m=1}^{k l_{\max}} a(m) D^{-m}$$

$l_{\max} = \max_i l_i$

where $l_{\max}$ is the maximum CW length (of the nonextended code C) and $a(m)$ is the number of source sequences x^k mapping into CWs of length m

- But the code is uniquely decodable, so there is at most one sequence mapping into each code m-sequence and there are at most D^m code m-sequences
- Thus, $a(m) \le D^m$, and we have
- Hence,

$$\left(\sum_{x \in \mathcal{X}} D^{-l(x)} \right)^k = \sum_{m=1}^{k l_{\max}} a(m) D^{-m} \le \sum_{m=1}^{k l_{\max}} D^m D^{-m} = k l_{\max}$$

$$\sum_i D^{-l_i} \le (k l_{\max})^{\frac{1}{k}}$$

This breaks for infinite cardinality of $\mathcal{X}$, since $l_{max}=\infty$

McMillan's Theorem (proof cont'd)

- Since this inequality is true for all k, it is true in the limit as $k \to \infty$
- Since $(kl_{\max})^{1/k} \to 1$, we prove the Kraft inequality

$$\sum_i D^{-l_i} \leq 1$$

$$\lim_{k\to\infty} (kl_{\max})^{\frac{1}{k}} = \lim_{k\to\infty} e^{\ln(kl_{\max})^{\frac{1}{k}}} = \lim_{k\to\infty} e^{\frac{1}{k}\ln(kl_{\max})}$$

$$= \lim_{k\to\infty} e^{\frac{\ln k + \ln l_{\max}}{k}} = e^{\lim_{k\to\infty} \frac{\ln k + \ln l_{\max}}{k}}$$

de l'Hôpital → $$= e^{\lim_{k\to\infty} \frac{\frac{1}{k}+0}{1}} = e^{\lim_{k\to\infty} \frac{1}{k}} = e^0 = 1$$

- Conversely, given any set of $l_1, l_2, \ldots, l_m$ satisfying the Kraft inequality, we can construct an instantaneous code
- Since every instantaneous code is uniquely decodable, we have also constructed a uniquely decodable code

Corollary

- A uniquely decodable code for an infinite source alphabet $\mathcal{X}$ also satisfies the Kraft inequality

  - *Proof:* The point at which the preceding proof breaks down for infinite $|\mathcal{X}|$ is [_], since for an infinite code l_{max} is infinite
  - But there is a simple fix to the proof
  - Any subset of a uniquely decodable code is also uniquely decodable
  - Any finite subset of the infinite set of CWs satisfies the Kraft inequality, so

$$\sum_{i=1}^{\infty} D^{-l_i} = \lim_{N\to\infty} \sum_{i=1}^{N} D^{-l_i} \leq 1$$

  - Given CW lengths $l_1, l_2, \ldots$ as in Kraft → construct an instantaneous code
  - Since instantaneous codes are uniquely decodable, we have a uniquely decodable code with an infinite number of CWs

- The McMillan theorem also applies to infinite alphabets

McMillan's Theorem Consequences

- The theorem implies a rather surprising result

  The class of uniquely decodable codes does not offer any further choices for the set of CW lengths than the class of prefix codes

- The set of achievable CW lengths is the same for uniquely decodable and instantaneous codes
- Hence, the bounds derived on the optimal CW lengths continue to hold even when we expand the class of allowed codes to the class of all uniquely decodable codes